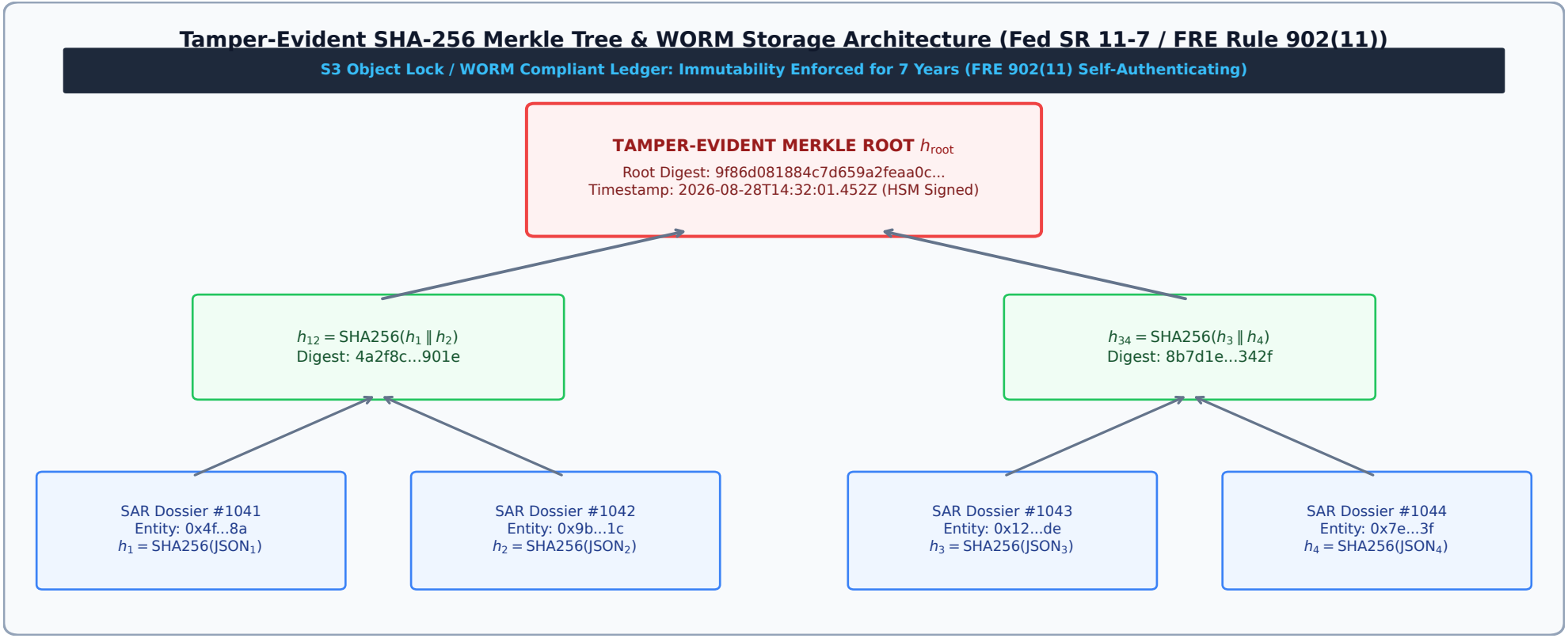
Tamper-Evident SHA-256 Merkle Tree & WORM Storage Architecture (Fed SR 11-7 / FRE Rule 902(11))
S3 Object Lock / WORM Compliant Ledger: Immutability Enforced for 7 Years (FRE 902(11) Self-Authenticating)
TAMPER-EVIDENT MERKLE ROOT h_{root}
Root Digest: 9f86d081884c7d659a2feaa0c...
Timestamp: 2026-08-28T14:32:01.452Z (HSM Signed)
$h_{12} = SHA256(h_1 \| h_2)$
Digest: 4a2f8c...901e
$h_{34} = SHA256(h_3 \| h_4)$
Digest: 8b7d1e...342f
SAR Dossier #1041
Entity: 0x4f...8a
$h_1 = SHA256(JSON_1)$
SAR Dossier #1042
Entity: 0x9b...1c
$h_2 = SHA256(JSON_2)$
SAR Dossier #1043
Entity: 0x12...de
$h_3 = SHA256(JSON_3)$
SAR Dossier #1044
Entity: 0x7e...3f
$h_4 = SHA256(JSON_4)$